

# Brain Tumor MRI Classification Using Convolutional Neural Networks

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**Abstract**—Brain tumor Diagnosis of brain tumor using magnetic resonance imaging (MRI). One of the important tasks in medical image is imaging (MRI) scan analysis. Manual reading of MRI images by radiologists may be the best option for evaluating this problem can take a long time and be inaccurate. This project proposes a Convolutional Neural Network (CNN) based brain tumor a classification model that can be used to classify magnetic resonance images into By classification, the types of brain tumors are divided into four classes: Glioma, Meningioma, Pituitary Tumor and No Tumor. The model is trained with a public available dataset and tested with publicly available Kaggle dataset. Experimental results show that CNN model classify, with high classification accuracy, and can assist to realize timely and effective. models can classify with high accuracy and can help realize timely and effective.

**Index Terms**—Brain Tumor, MRI, CNN, Deep Learning, Medical Imaging, Classification

## I. INTRODUCTION

Brain tumors are some of the most severe brain conditions. problems impacting people's health. To make early and accurate diagnosis is necessary for the effective treatment planning and better outcomes. Magnetic resonance imaging (MRI) Offering ultimate low-light sensitivity, the cholesteric liquid crystal provides highly resolved images of is widely used brain tissues. Diagnosis is traditionally heavily dependent on radiologists and radiology .An examination of an MRI image can be done manually. Recent advancements in based on Artificial Intelligence (AI) and Deep Learning, AI Based Weather Prediction has been made possible. Brain Tumor automatic detection and classification. Convolutional Neural Network(CNN) one of the most popular network tools, albeit one with a potentially quite restricted function:convolutional Neural Networks (CNNs).The most successful deep learning models that are suitable for image analysis tasks classification. This project is mainly dedicated to the development of the CNN based model Can be used to classify brain MRI automatically into different tumor categories.

## II. DATASET

Brain Tumor MRI Dataset downloaded from Kaggle with MRI images of Glioma, Meningioma, Pituitary Tumor, and No Training and testing CNN model classes arrayed in a 2D array.

## III. RELATED WORK

There have been a number of researchers who have used machine learning and deep learning techniques for brain tumor classification using learning techniques. Traditional Two hand-crafted image features and methods were used quite a number of sifers, including the Support Vector Machines (SVMs). More recent research has demonstrated that CNN based techniques are used to effectively recognize handwritten digits.outperform the traditional techniques by automatically extracting, Use Magnetic Resonance Imaging (MRI) to detect discriminative features in the image. Transfer learning In particular, architectures like ResNet, VGG16, and EfficientNet have been considered. The same has also been applied to medical image classification problems. Despite their simplicity and effectiveness, CNN models are still popular. Approach to ability to learn hierarchical image representa- tions.

## IV. DATASET DESCRIPTION

This project uses the data set of the Brain Tumor. This MRI Dataset has been provided by Kaggle. Includes MRI images This was determined and categorized into four classes:

- Glioma Tumor
- Meningioma Tumor
- Pituitary Tumor
- No Tumor

The dataset includes separate training and testing folders with labeled MRI images.

### **Dataset URL:**

<https://www.kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset>

## V. PROBLEM STATEMENT

The process of manual diagnosis of brain tumors from MRI images is time consuming. Consumption is the consumption and it's a costly procedure that is only suitable for very skilled radiologists. Human errors Diagnosis can be affected by rators and the inter-observer variability may affect diagnosios accuracy. The aim of this study is to create an automation system for Portrait of a brain tumor classification model using CNN.Efficient CNN-based brain tumor classification model with high accuracy.Produce unreliable categorisation of tumour types using MRI images

## VI. RESEARCH QUESTIONS

- RQ1: Can CNN models accurately classify brain tumor MRI images?
- RQ2: Does data augmentation improve CNN classification performance?
- RQ3: How does model complexity influence classification accuracy?
- RQ4: What are the common causes of misclassification?
- RQ5: How does dropout regularization affect model performance?

## VII. METHODOLOGY

The proposed methodology consists of the following stages:

- 1) Data Collection
- 2) Data Preprocessing
- 3) Image Resizing and Normalization
- 4) CNN Model Design
- 5) Model Training
- 6) Model Evaluation
- 7) Result Analysis

The images in MRI underwent the standardization of size and normalization to 224×224 pixels. The MRI images were resized and standardised to 224×224 pixels before training. Data augment techniques include rotations, etc., Some horizontal squaring was used, as well as some 'zooming' to enhance the quality of some of the photos.

## VIII. EXPERIMENTAL SETUP

Experiments were conducted using Python, TensorFlow, and Keras on the Kaggle platform.

The CNN architecture consists of:

- Convolution Layers
- ReLU Activation Functions
- Max Pooling Layers
- Fully Connected Dense Layers
- Dropout Layer
- Softmax Output Layer

The Adam optimizer and categorical Cross-Entropy function. The function was employed in the learning process

## IX. MODEL ARCHITECTURE

The suggested CNN proposed model is the classification of the brain. There are four types of MRI images provided: Glioma Tumor, Meningioma Tumour, pituitary tumour, and No Tumor. The architecture consists of several layers extracting image properties.

The CNN architecture includes:

- Input Layer ( $224 \times 224 \times 3$  MRI images)
- Convolution Layer with ReLU activation
- Max Pooling Layer
- Second Convolution Layer with ReLU activation
- Max Pooling Layer
- Third Convolution Layer with ReLU activation
- Max Pooling Layer
- Flatten Layer

- Fully Connected Dense Layer
- Dropout Layer
- Softmax Output Layer (4 Classes)

The spatial features are automatically learned by the convolution layers. Layering splits data to gain dimensions; from MRI images. Layering decreases dimensionality from MRI images and computational complexity.

## X. RESULTS AND DISCUSSION

The CNN model achieved good results on the testing dataset. We noticed that the data within the model were still sufficient for improving the accuracy of validation and reduced overfitting. To create more complex models, any additional convolution layers other than the basic two can be added for accuracy than any other previous studies.

A dropout was added on top of the regularization that resulted in better generalization reduced training bias. Misclassification analysis showed that Some types of tumors have identical features, and makes classification more difficult. The confusion matrix showed that most of the images were classified correctly. The correct classification of images was achieved

## XI. RESEARCH QUESTION ANALYSIS

### A. RQ1

CNN models successfully classified MRI brain tumor images with high accuracy, demonstrating their effectiveness for medical image analysis.

### B. RQ2

Data augmentation improved model robustness and helped prevent overfitting by increasing image diversity.

### C. RQ3

Increasing model complexity improved feature extraction capabilities but also increased computational cost.

### D. RQ4

Misclassifications mainly occurred due to similarities between tumor structures and image quality variations.

### E. RQ5

Dropout regularization improved validation performance and enhanced model generalization.

## XII. OUTPUT RESULTS

RQ1: Can CNN models accurately classify brain tumor MRI images?

The CNN model was trained for 3 epochs with the Brain Tumor MRI dataset. When tested, model hit an accuracy of categories. obtained an accuracy of 81.56% and was able to successfully categorize MRI images into 4 categories

### RQ2: Does Data Augmentation Improve CNN Classification Performance?

Two CNN models were trained and compared: one without data augmentation and one with data augmentation. The model without augmentation achieved a validation accuracy of 86.34%, while the model with augmentation achieved a validation accuracy of 61.87%.

| Model                    | Accuracy (%) |
|--------------------------|--------------|
| CNN Without Augmentation | 78.38        |
| CNN With Augmentation    | 65.25        |

TABLE I  
COMPARISON OF CNN MODELS WITH AND WITHOUT DATA AUGMENTATION

The results indicate that the CNN model without augmentation performed better, achieving an accuracy of 78.38% compared to 65.25% for the augmented model. This may be due to the limited training epochs and the increased complexity introduced by augmentation. Therefore, data augmentation did not improve classification performance in this experiment.

### RQ3: How Does Model Complexity Influence Classification Accuracy?

Three CNN architectures with different levels of complexity were evaluated: Simple CNN, Medium CNN, and Deep CNN. The objective was to analyze the impact of increasing model complexity on brain tumor classification performance.

| Model      | Accuracy (%) |
|------------|--------------|
| Simple CNN | 80.69        |
| Medium CNN | 80.50        |
| Deep CNN   | 77.94        |

TABLE II  
PERFORMANCE COMPARISON OF CNN ARCHITECTURES

The highest accuracy of 80.69% was obtained by the Simple CNN. The second most frequent was followed closely by the Medium CNN with 80.50%. The Deep lowest accuracy of 77.94% was obtained by CNN. The results show that complexity of the model gave no benefits. classification performance. Other, more complex, architectures will also be considered need more epochs/training time and more computational resources to ensure they get a better outcome

## XIII. CONCLUSION

This was a Basic project of brain tumor diagnosis using an MRI image and CNN.classification system that can classify MRI images into four categories. With experimental results showing that CNN models are able to deliver consistent classification accuracy and can help to support automated diagnosis systems. It is possible that the transfer learning architectures will be used in future work. Extensive approaches like ResNet50, EfficientNet and Vision Transformers to further improve in the classification accuracy

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